# Akshay Bhat

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GitHub Profile | LinkedIn Profile | Portfolio Website

## Education

**B.M.S College Of Engineering, Bengaluru** *2024 – 2028*

Bachelor of Engineering, Artificial Intelligence and Machine Learning CGPA: 9.09

## Technical Skills

**-Languages**: C++ | Python | C | SQL | HTML | CSS
**-Frameworks**: React.js | Next.js | Tailwind | Flask | PyTorch | Scikit-learn
**-ML/AI**: Computer Vision | NLP | Classical ML | Transformers | RAG & Vector DBs | Explainable AI
**-Tools**: Git | Google Colab | GCP | Jupyter Notebook | MongoDB | MySQL | SQLite | Vercel | Render
**-Course Work**: Data Structures | Databases | OOP | Computer Architecture | Machine Learning

## Experience

**Hitachi** *Aug 2026 – Sep 2026*

Research and Development Intern WTC, Bengaluru

- Recreated the NVIDIA Omniverse digital twin pipeline in **PyBullet** with injected motor faults.
- Trained an **LSTM Autoencoder** on FFT features, using ONNX Runtime and EWMA for anomaly detection.
- Built a **FastAPI** backend with WebSockets and MongoDB for a Flutter and Three.js dashboard.
- Implemented **Flutter** anomaly-monitoring screens from provided designs, including floor plans and visualizations.

**Delivery Hub LLC – ParcelVision** *Feb 2026 – May 2026*

Computer Vision & Research Intern Bengaluru

- Built a 3D object measurement pipeline using **YOLOv8-World**, **SAM**, and geometric reconstruction.
- Implemented **IMU fusion**, optical flow, and **Structure from Motion** for monocular scale recovery.
- Built a 3D point cloud reconstruction pipeline using camera intrinsics and depth back-projection.
- Evaluated **DepthAnything V2** and **ZoeDepth**, identifying **ARCore/LiDAR** as the practical metric-scale solution.

## Personal Projects

**ContractPulse** | *Python, RoBERTa, DistilBERT, Prophet, Flask, Next.js, MongoDB* *Apr 2026*

Fintech SaaS platform for AI-powered legal contract analysis and breach risk forecasting.

- Fine-tuned **RoBERTa QA** on **CUAD**, extracting covenants as structured JSON with confidence scores.
- Built a multi-stage NLP pipeline with **DistilBERT NLI** and Groq APIs for semantic conflict detection.
- Containerized and deployed the Flask + Next.js platform on **Hugging Face Spaces** and **Vercel**.

**NeoScore** | *Python, XGBoost, SHAP, Flask, Next.js, MongoDB* *Mar 2026*

Explainable AI credit scoring platform for thin-file users in the Indian lending ecosystem.

- Trained a teacher–student **XGBoost** pipeline on **307K+ records** via knowledge distillation.
- Engineered 30+ features and calibrated probabilities using **isotonic regression**, achieving **ROC-AUC 0.79**.
- Built a **SHAP** + **Groq**-based explainability pipeline for personalized credit recommendations.

**MoodQuest** | *PyTorch, OpenCV, Flask, Next.js, MongoDB, Docker* *Nov 2025*

Real-time facial expression recognition and mental health monitoring platform.

- Built a real-time inference pipeline combining **OpenCV** face detection with an **EfficientNet-B0** emotion classifier.
- Built a Flask backend powering **PHQ-9/GAD-7/PSS** scoring, mood analytics, and a **Groq**-based chatbot.
- Containerized and deployed the backend on **HF Spaces** via **Docker**, with **MongoDB** persistence.

## Positions of Responsibility

**Web Developer & UI Designer** *Sept 2025 – Feb 2026*

AeroBMSCE Club

## Achievements

- Top 6 of 100 teams — AiVerse 2.0 Hackathon *Oct 2025* — **MoodQuest**
- Top 15 of 125 teams — Phaseshift Tech Fest '25 *Sept 2025* — **CareBridgeAI**